Scope of Work (SOW)

Team Member(s): Gabriel Goeken
Date submitted for review: January 29, 2026
Submitted to: Civil and Environmental Engineering Department faculty

Project Goals: This project aims to perform a replicable and technically justifiable analysis of AirPurple air-quality monitoring data collected across Nebraska from February 2024 through March 2025. That analysis will characterize spatial and temporal air-quality conditions, evaluate compliance with EPA National Ambient Air Quality Standards (NAAQS) and Air Quality Index (AQI) thresholds, identify potential air-quality hotspots relevant to public health, and assess the influence of meteorological and geographic factors on pollutant concentrations.

Project Work: Task 1: Data Acquisition and Preparation. Obtain and organize all the AirPurple monitoring data and sensor metadata. Check the structure and units, timestamps, and spatial attributes of the dataset. Create a replicable directory and version control. Task 2: Data Cleaning and Quality Control. Import data into Python for analysis. Check all missing, duplicate, or invalid values and correct them. Test for anomalous sensor readings using statistical thresholds. Record all quality control measures. Task 3: Data Structuring and Categorization. Categorize all records by sensor name for location-specific analysis. Sort temperature and relative humidity into bins. Analyze elevation and geographic information in the data. Task 4: Statistical and Spatial Analysis. Calculate mean, median, and maximum levels of VOC, $PM_{2.5}$, and PM_{10} by sensor—rank sensor locations to discern elevated concentration trends. Identify dates and locations associated with peak pollutant values. Task 5: Temporal, Meteorological, and AQI Evaluation. Evaluate pollutant patterns over time and across meteorological categories. Convert $PM_{2.5}$ and PM_{10} concentrations to AQI values using EPA breakpoints. Determine and assess periods that pose AQI health risks. Explore and understand elevated pollution events. Task 6: Exploratory Altitude Assessment (Bonus). Examine the relationship between sensor elevation and pollutant concentrations. Record observed trends and limitations of the analyses used.

Project Deliverables:

Deliverable:	Description:	Due Date:
Raw Data Files	CSV files used to execute the Python analysis	2/3/26
Python Code Notebook	Fully documented Jupyter Notebook (.ipynb)	2/3/26
GitHub Repository	Public repository with code, data, and README	2/3/26
Annotated Code Document	Line-by-line explanation of Python code	2/3/26
Technical Report	Written report with background, methods, results, and references	2/3/26
Final Scope of Work	Completed SOW defining tasks and responsibilities	2/3/26